

BAYESIAN APPLICATIONS IN TIME SERIES

Jan T. Grevet

†Department of Statistical Science, Duke University

Duke

Bayesian Bivariate Markov Switching Multifractal Model

Introduction

Markov Switching Multifractal Model (MSM) is a stochastic volatility model proposed by Calvet and Fisher (2006)[1]. It models the volatility of financial series as a realization from a zero-mean Normal distribution with the variance being a product of several first-order two-state Markov switching processes with varying switching probabilities. Intuitively, it can be understood as a probabilistic and product based equivalent of Fourier transform which models time series with a sum of deterministic harmonic functions.

The strength of this methodology is that similar to Fourier transform, it allows decomposition of univariate series into several probabilistic Markov switching processes. Consequently, it provides natural characterization of financial shocks as an aggregate of stochastic regime switches with different frequency of occurrences.

Our proposed Bayesian Bivariate Markov Switching Multifractal Model funded by BEST award grant and mentored by Prof. Ma, generalizes the Bivariate MSM of Calvet and Fisher (2006)[1] by rephrasing it in Bayesian format and relaxing the constraint by allowing all combination of decomposed Markov switching processes across series to potentially have conditional dependence.

Model Specification

Univariate MSM requires 4 parameters $\{m_0, \gamma_1, b, \bar{\sigma}\}$ to be estimated (or sampled from posterior in our case) and another model parameter $\bar{k}$ to be set by user. Description of these parameters are as follows:

- $\bar{k}$: Number of Markov switching components to be decomposed into.
- m_0 : The state value of each Markov switching processes (the other state value when this is set is given deterministically as $m_1 = 2 - m_0$)
- γ_1 : Switching probability of the first layer Markov switching process.
- b : A parameter when coupled with γ_1 defines switching probability of 2nd to $\bar{k}$ th Markov switching processes by $\gamma_i = 1 - (1 - \gamma_1)^{(b^i - 1)} \forall i$.
- $\bar{\sigma}$: Scaling parameter of the volatility.

The hierarchical representation of this univariate model is the following:

$$\begin{aligned} r_t &\sim N(0, \bar{\sigma}^2 \prod_{i=1}^{\bar{k}} M_{i,t}), & r_t : \text{log difference of returns} \\ M_{i,t} &= M_{i,t-1} + \mathbb{1}_{u < \gamma_i}(\bar{m} - M_{i,t-1}), & u \sim U(0, 1) \\ \bar{m} &\sim m_0 \mathbb{1}_{u' < 0.5} + (2 - m_0) \mathbb{1}_{u' \geq 0.5}, & u' \sim U(0, 1) \\ \gamma_i &= 1 - (1 - \gamma_1)^{(b^i - 1)}, & i = 1, \dots, \bar{k} \end{aligned}$$

For our bivariate case where we have two series α, β , there are pairs of these parameters $\{(m_0^\alpha, m_0^\beta), (\gamma_1^\alpha, \gamma_1^\beta), (b^\alpha, b^\beta), (\bar{\sigma}_\alpha, \bar{\sigma}_\beta)\}, (\bar{k}^\alpha, \bar{k}^\beta)$ as well as correlation ρ and latent variables called switches $1_{1:\bar{k}^\alpha, t}^\alpha \in \{0, 1\}^{\bar{k}^\alpha}, 1_{1:\bar{k}^\beta, t}^\beta \in \{0, 1\}^{\bar{k}^\beta}$ and coefficient for these switches $B_{1:\bar{k}^\alpha, 1:\bar{k}^\beta} \in \{0, 1\}^{(\bar{k}^\alpha \times \bar{k}^\beta)}$. Then, by specifying conditional dynamics of α given β by regression, we have a joint model for bivariate returns as follows:

$$\begin{aligned} \begin{pmatrix} r_t^\alpha \\ r_t^\beta \end{pmatrix} &\sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha & \rho \sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha} \sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} \\ \rho \sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha} \sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} & \bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta \end{pmatrix} \right] \\ M_{i,t}^\beta &= M_{i,t-1}^\beta + 1_{i,t}^\beta (\bar{m}^\beta - M_{i,t-1}^\beta), \quad i = 1, \dots, \bar{k}^\beta, \quad 1_{i,t}^\beta \sim \text{Bernoulli}(\gamma_i^\beta) \\ M_{i,t}^\alpha &= M_{i,t-1}^\alpha + 1_{i,t}^\alpha (\bar{m}^\alpha - M_{i,t-1}^\alpha), \quad i = 1, \dots, \bar{k}^\alpha, \quad 1_{i,t}^\alpha \sim \text{Bernoulli}(\pi_i) \\ \text{logit}(\pi_i) &= C + B_{i,1} 1_{1,t}^\beta + \dots + B_{i,\bar{k}^\beta} 1_{\bar{k}^\beta, t}^\beta, \quad i = 1, \dots, \bar{k}^\alpha, \quad C = \text{logit}(\gamma_i^\alpha) - B_{i,1} \gamma_1^\beta - \dots - B_{i,\bar{k}^\beta} \gamma_{\bar{k}^\beta}^\beta \end{aligned}$$

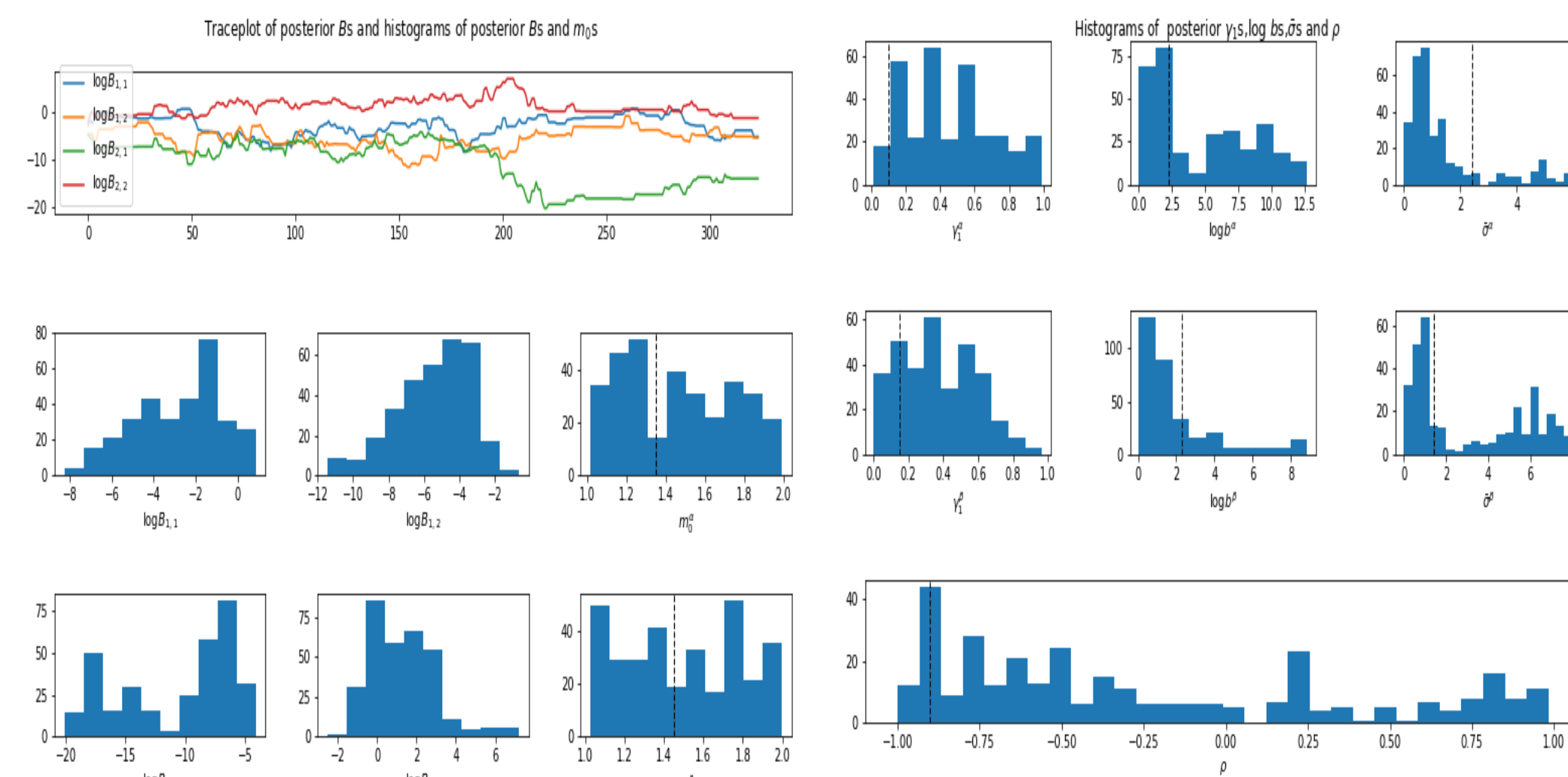
where we can trivially check that π_i when marginalizing out $1_{1:\bar{k}^\beta, t}^\beta$ is γ_i^α .

Sampling Algorithm

Now that we have a full model specification, and since state space is finite and probability of all possible transitions can be computed exactly in efficient manners by distributed computation (here we use Spark), we have access to the exact joint likelihood (the form of it is given in my teammate Jaeyeon Lee's poster) of these parameters marginalizing out latent variables such as switches. Then, the sampling algorithm proceeds like this:

1. Pick appropriate $\bar{k}^{\alpha, \beta}$ as model parameters.
2. Initialize $m_0^{\alpha, \beta}, \gamma_1^{\alpha, \beta}, b^{\alpha, \beta}, \bar{\sigma}^{\alpha, \beta}, B_{1:\bar{k}^\alpha, 1:\bar{k}^\beta}$
3. Set priors on parameters: $Beta()$ for γ_1 s, $Gamma()$ for b s and $Exp()$ for B s (to induce shrinkage and we know $B_{i,j} > 0 \forall i, j$), improper priors for others.
4. Sample B s from normal proposal and accept it with the Metropolis acceptance rate
5. In case sampled B s change the strict ordering of π_i s ($\pi_1 < \pi_2 < \dots < \pi_{\bar{k}}$), reject proposed B s.
6. Sample other parameters with Metropolis reflecting random walk (to account for its domain), and compute the acceptance probability of these new set of parameters and accept/reject with that rate.
7. Back to 4 until good mixing (for better mixing performance, we can make step 4 and 6 into multiple steps as well).

Results& Comments



Here, we simulated synthetic data that follows MSM with $\bar{k} = (2, 2)$ and only $B_{2,1}$ being nonzero, meaning 2nd layer Markov Switching Processes of α is conditionally dependent on switches in 1st layer of β series.

Resulting posterior draws from our MCMC suggest some mixing issues. The traceplot of $\log b$ s (not shown here) shows that parameter space of b s are not explored enough even though we tried several proposal distributions (reflecting uniform, log-normal with Jacobian corrections etc.). Thus, more efficient MCMC such as HMC or better computing environment (with a lot of cores to distribute jobs) may be needed.

Still, posterior distributions of some parameters are already concentrated around the true value (horizontal dotted line for each histogram) and we can also observe the multi-modal nature of some parameters which is theoretically given (that may be the source of mixing problem).

The interesting part is that although the result of the traceplot of B s do suggest that there are only one non-zero components (remember that it's in log scale), it somehow recovered $B_{2,2}$ to be that component rather than $B_{2,1}$ in the true model. Thus it did manage to find that second layer of α is dependent on switches in one layer of β but missed the specific layer.

Model Based Missing Value Imputation in Time Series

Introduction

This project undertaken in STA642 course combines several methodologies learned in the class to do model based missing value imputation on time series data with the assumption that the data are missing completely at random, thus a model based extrapolation from the observed data would suffice.

Specifically, the problem this algorithm addresses is in the case of constructing Dynamic regression and both predictors and responses are missing in some extent. The data I use to visually evaluate the performance of the resulting methodology is salesindex.mat data from STA642 course which contains two series Sales and Index where we use Index to predict Sales.

Methodology

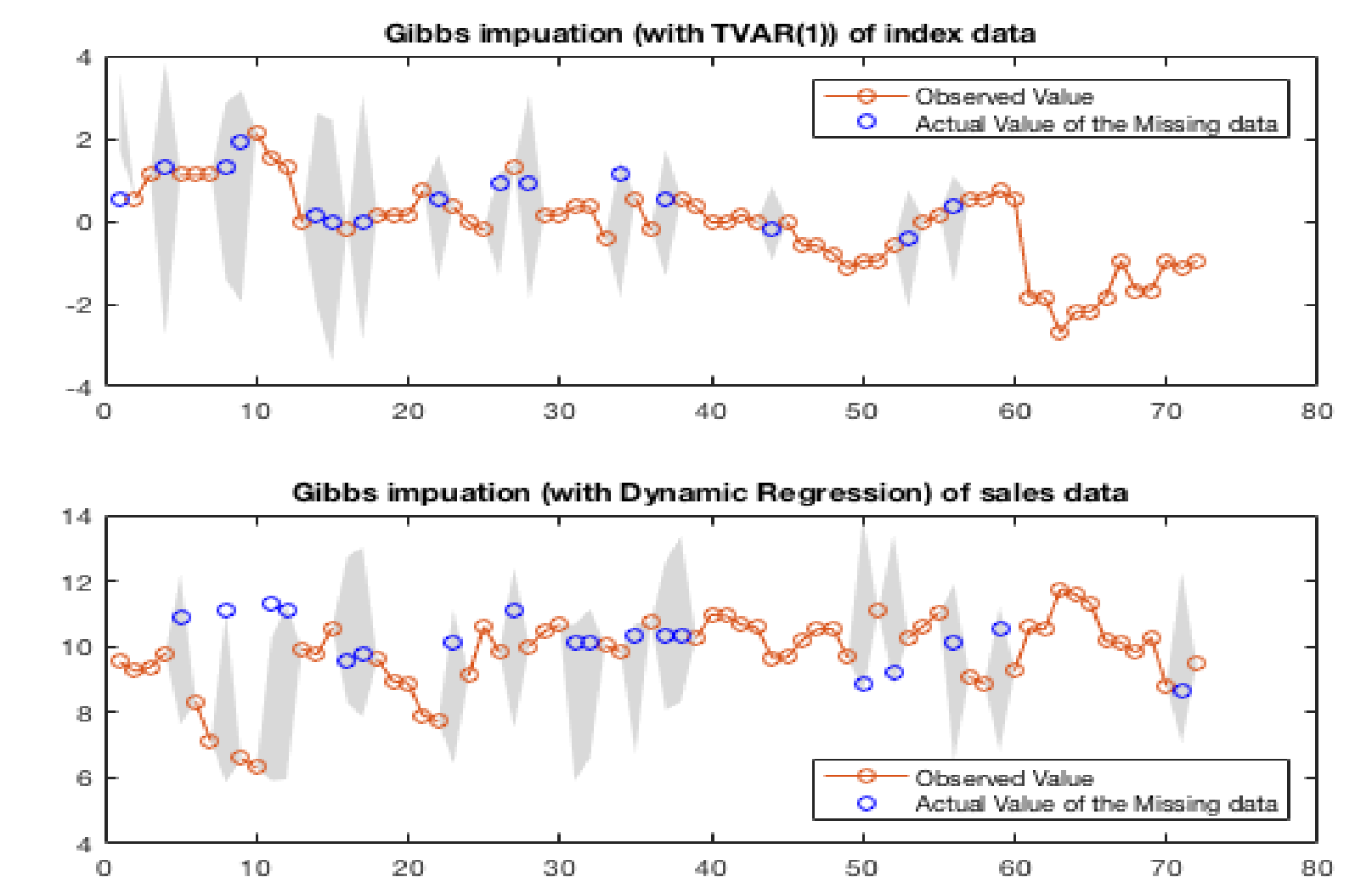
The imputation algorithm is constructed by two Gibbs samplers as given below:

Algorithm 1 Predictor Imputation	Algorithm 2 Response imputation
procedure GIBBS FOR PREDICTOR $X \equiv \{X^{\text{OBS}}, X^{\text{MISS}}\}$ Set prior, discount and model params ϕ Initialize X_0^{MISS} for $s = 1$: number of Gibbs runs (=S) do $X_s^{\text{OBS}} \sim X_{s-1}^{\text{MISS}}, \phi$ $\triangleright$ by TVAR(p) $X_s^{\text{MISS}} \sim X_s^{\text{OBS}}, \phi$ $\triangleright$ Del & Imp return $X_s \equiv \{X_s^{\text{OBS}}, X_s^{\text{MISS}}\} \forall s \in S$	procedure GIBBS FOR RESPONSE $Y \equiv \{Y^{\text{OBS}}, Y^{\text{MISS}}\}$ AND PARAMS Run Algorithm 1, get $X_s(T \times S)$ Set prior, discount and model params ψ Initialize Y_0^{MISS} for $s = 1$: S do $Y_s^{\text{OBS}} \sim Y_{s-1}^{\text{MISS}}, X_s, \psi$ $\triangleright$ Dynamic Reg $Y_s^{\text{MISS}} \sim Y_s^{\text{OBS}}, X_s, \psi$ $\triangleright$ Del & Imp return $Y_s \equiv \{Y_s^{\text{OBS}}, Y_s^{\text{MISS}}\} \forall s \in S$ & Posterior of Coefficients etc.

Where Del & Imp stands for the deletion of observations (previously imputed missing values) to update state estimates of remaining observations followed by imputation of these deleted data as in Harrison and Veerapen (1993)[2].

Results & Comments

Resulting bounds for missing data compared with the actual values of these are plotted below for both predictor series Index and response series Sales.



Even though I deleted around 20% of the observations, bounds for Index series are reasonably narrow especially for data around the middle which has a lot of observations before and after to be utilized in forward and backward sampling. For response series, since uncertainty from imputed predictors and response both feeds in, bounds are slightly worse.

References

- [1] Laurent E Calvet, Adlai J Fisher, and Samuel B Thompson. "Volatility comovement: a multifrequency approach". In: *Journal of econometrics* 131:1-2 (2006), pp. 179-215.
- [2] P.J Harrison and P.P Veerapen. "Incorporating and deleting information in dynamic models". In: *Developments in Time Series Analysis: in Honour of Maurice B. Priestley*. Chapman and Hall London, 1993, pp. 37-49.